Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Variance Decomposition Across Five Engineering Choices

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Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. We hold a single spur gear pair (module 3 mm,20 teeth,42CrMo4/AISI 4140,700 N·m,constant amplitude, $R = 0$)choices: size-and-surface standard (ISO 6336, AGMA 2001, FKM), surface roughness grade (Rz = 4, 15, 50 μm), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336 Y_{Sa} factor within the nominal stress calculation; a separate fatigue notch factor K_f was removed to avoid double-counting of the notch effect. A full-factorial ANOVA over 144 combinations28 run-outssize-and-surface standard accounts for 31.1% of variance, statistically indistinguishable from mean-stress correction (30.9%). Surface roughness contributes 8.5%, with a Standard × Rz interaction of 5.0%. The S–N curve source (0.7%) and cumulative damage rule (0.5%) are minor. Bootstrap validation (2000 drawsstable rankings. The factor ranking is operating-point-dependent: at industrially common torque levels the standard choice dominates, while mean-stress correction rises to co-equal status only near the endurance-limit-proximity regime explored here. Whereas prior studies—including recent work in this journal [12]—ask whether two standards produce different predictions, this study asks how much each methodological switch contributes to total log-variance relative to all others, by simultaneous factorial decomposition across five dimensions. No prior work has ranked the sources of method-induced divergence in gear fatigue life prediction by variance explained. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dexagainst the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

Keywords: gear fatigue, uncertainty quantification, ANOVA, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear's fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

The variance-decomposition methodology is a standard tool for identifying dominant sources of divergence when different analysis pipelines are applied to identical input data. Here, we apply this framework to gear fatigue life prediction.

We address two questions: which methodological choices produce the largest divergence in predicted gear bending fatigue life, and can a bootstrap noise floor distinguish genuine signal from finite-sample fluctuation?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ultimate tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

2.2 Five Methodological Switches

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation $\sigma_a = \sigma'_f \cdot (2N_f)^b$ relates stress amplitude to cycles to failure. We use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 ($\sim$300 HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

The 328 MPa spread in σ'_f and the 50 MPa difference in endurance limits reflect genuine test-to-test variability across different laboratories and specimen batches for the same nominal material condition. The raw data are freely downloadable; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma_{\text{uts}})$
- **Gerber:** $\sigma_{ar} = \sigma_a/(1 - (\sigma_m/\sigma_{\text{uts}})^2)$
- **Morrow:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma'_f)$, where σ'_f is taken from the active S–N curve
- **SWT:** $\sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a}$

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R\,\text{rel}\,\text{T}} = 1.674 - 0.529(R_z+1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R\,\text{rel}\,\text{T}} = 1.120$ for $R_z < 1$ μm.
- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch $= 8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \to R_a$ via $R_a \approx R_z/6$.
- **FKM Guideline (7th ed.):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}), 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4 μm:** Precision aerospace quality
- **Machined, Rz = 15 μm:** Standard industrial quality
- **As-forged, Rz = 50 μm:** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 1) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta\,\mathrm{rel\,T}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta\,\mathrm{rel\,T}}$ is not among our switches, we omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa}.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa}\, Y_{Sa}\, Y_\varepsilon\, Y_\beta \tag{1}$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700$ N·m, $d_1 = m_n z_1 = 60$ mm, giving $F_t \approx 23.3$ kN); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20°$, standard ISO 53 basic rack, no profile shift); $Y_\varepsilon = 0.25 + 0.75/\varepsilon_\alpha \approx 0.691$ with $\varepsilon_\alpha \approx 1.7$; and $Y_\beta = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778$ MPa.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A, dynamic factor K_V, face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B, and deep tooth factor Y_{DT}. All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are appropriate for a methodology-comparison study (introducing application-specific K_A and K_V would add noise without changing the relative ranking of the methodological switches) but constrain the absolute N_f values to an idealized gear pair.

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 1)

2. Apply mean stress correction (Switch 2) → σ_{ar}
3. Apply size and surface factors (Switch 4, Switch 5) → σ_{eff}
4. Solve Basquin equation (Switch 1) → N_f
5. Apply damage rule threshold (Switch 3) → $N_f^{(\text{final})}$

2.7 Statistical Analysis

2.7.1 Full-Factorial ANOVA

We perform a Type-II analysis of variance on $\log_{10}(N_f)$ with all five factors as categorical predictors. Because the design is balanced, Type-I, Type-II, and Type-III sums of squares are identical for main effects. The variance fraction attributed to factor j is $\text{SS}_j/\text{SS}_{\text{total}} \times 100\%$. We decompose the Standard × Rz interaction via the cell-means approach. All p-values serve as effect-size descriptors in the $N = 116$ regime, not as confirmatory tests.

2.7.2 Bootstrap Noise Floor

We bootstrap-resample the 116 finite-life entries 2000 times (RNGseed 42), recomputing all variance fractions on each draw. The signal-to-noise ratio $\text{S/N} = \bar{\theta}_j/\sigma_{\text{boot,j}}$ identifies factors robustly above sampling noise; a factor with S/N < 3 is indistinguishable from zero at the 95% confidence level.

3 Results

Table 2: Full-factorial ANOVA results ($2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations, 116 finite-life entries, 28 run-outs,$\sigma_{F0} \approx 778$ MPa). All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase. Bootstrap: 2000 draws, RNG seed 42.

Factor	SS	df	F	p	Var.%	S/N
Size/surface standard	40.2	2	67.9	$< 10^{-15}$	31.1	5
Mean stress correction	39.9	3	45.0	$< 10^{-15}$	30.9	5
Surface roughness, Rz	11.0	2	18.6	$< 10^{-15}$	8.5	2
Standard × Rz	6.5	4	5.5	0.0005	5.0	2
S–N curve source	0.9	1	2.9	0.089	0.7	1
Cumulative damage rule	0.7	1	2.4	0.128	0.5	1
Residual	30.1	102	—	—	23.3	—
Total	129.3	115	—	—	—	—

3.1 Variance Decomposition

Table 2 presents the ANOVA decomposition. Two factors dominate: the choice of size-and-surface standard (31.1%, S/N = 5) and mean-stress correction (30.9%, S/N = 4). Their bootstrap confidence intervals overlap substantially (standard: 95% CI [21.1, 44.9];

mean-stress: [18.6, 47.6]), making the ranking between them statistically ambiguous. Together they account for 62.0% of total log-variance.

Surface roughness (8.5%, S/N = 2) and the Standard × Rz interaction (5.0%, S/N = 2) are the only other factors distinguishable from zero, though their S/N ratios fall below the conventional threshold of 3. All other two-way and higher-order interactions were tested; none exceeded 2% of total variance. The S–N curve source (0.7%, $p = 0.089$) and cumulative damage rule (0.5%, $p = 0.128$) are not significant at the 5% level.

The residual (23.3%) reflects the nonlinearity introduced by the endurance-limit cut-off: 28 of 144 combinations predict run-out, and the binary finite-life/run-out transition creates variance that an additive main-effects model cannot fully capture.

3.2 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 458$ MPa), Goodman the highest (≈ 636 MPa), with Morrow (≈ 545 MPa) and SWT (≈ 550 MPa) intermediate. This 39% spread in equivalent stress translates to a factor of $\sim$10–100 in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\text{uts}}$ ratio is $\sim$0.39, large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible.

We emphasize that this finding applies to quenched-and-tempered steel without residual stresses. Case-hardened gears with compressive residual stresses would experience a lower effective stress ratio, reducing the mean-stress contribution.

3.3 Surface Roughness and Standard Effects

Surface roughness contributes 8.5% of variance. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40$ μm; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard × Rz interaction (5.0%) reflects the fact that the three standards' predictions diverge more at rough surface finishes than at ground finishes.

3.4 Minor Factors

The S–N curve source (0.7%) is negligible—the two Waterloo datasets, despite a 50 MPa endurance-limit difference, produce similar rankings of methodological choices. The cumulative damage rule (0.5%) is likewise negligible for constant-amplitude loading, where Miner vs. Modified Miner merely scales the life by a constant factor of 0.7.

3.5 Torque Dependence of the Factor Ranking

To test whether the factor ranking is an artifact of the chosen torque, we repeated the full 144-combination sweep at three torque levels: $T = 500$, 600, and 700 N·m (producing $\sigma_{F0} \approx 556$, 667, and 778 MPa, respectively). At 500 N·m, only 14 combinations predicted finite life—the ANOVA is severely underpowered (5 factors plus interaction on 14 observations) and the residual sum of squares becomes negative, confirming that the model is not identified at this torque level. We therefore report only the 600 and 700 N·m results:

Factor	600 N·m (60 finite)	700 N·m (116 finite)	Trend
Size/surface standard	36.0%	31.1%	stable
Standard × Rz	17.8%	5.0%	decreasing
Mean stress correction	10.2%	30.9%	**increasing**
S–N curve source	7.9%	0.7%	decreasing
Surface roughness, Rz	5.8%	8.5%	modest
Cumulative damage	1.1%	0.5%	negligible

The mean-stress correction contribution more than triples when torque rises from 600 to 700 N·m. This is physically expected: the $\sigma_m/\sigma_{\text{uts}}$ ratio increases from ∼0.33 to ∼0.39, and the functional-form differences among Goodman, Gerber, Morrow, and SWT grow nonlinearly with stress. At lower, industrially common torque levels, the standard choice dominates; at higher torque, mean-stress correction becomes co-equal. The factor ranking is therefore *not* a universal property of gear fatigue methodology— it depends on the operating point. We recommend that future extensions of this work characterize the full torque–ranking response surface.

3.6 Fatigue Life Spread

Across all 144 combinations, $\log_{10}(N_f)$ ranges from 2.52 to 7.50among the 116 finite-life entries (3.3×10^2 to 3.1×10^7 cycles, 5.0 dex). The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50$ μm); the longest to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4$ μm). The 28 run-out combinations predominantly occur with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve under the Gerber correction.

3.7 Bootstrap Stability

Bootstrap 95% confidence intervals confirm the top-two factor ranking is stable under resampling.[18.9, 44.2]) and mean-stress correction (mean 32.4%, CI [19.9, 45.3]) have overlapping CIs. Surface roughness (mean 9.9%, CI [2.3, 20.9]) and the interaction term (mean 6.2%, CI [0.0, 14.0]) are distinguishable from zero but with wider relative uncertainty. The S–N source (mean 1.5%, CI [0.0, 6.4]) and damage rule (mean 1.4%, CI [0.0, 6.2]) have CIs that include zero.

4 Discussion

4.1 Two Dominant Factors, Statistically Indistinguishable

The central result is that two factors account for 62.0% of total log-variance and cannot be statistically separated: the choice of standard (ISO 6336, AGMA 2001, FKM) and the choice of mean-stress correction (Goodman, Gerber, Morrow, SWT). A practicing engineer who selects the most conservative combination (FKM + Goodman) will predict a fatigue life two orders of magnitude shorter than one who selects the least conservative

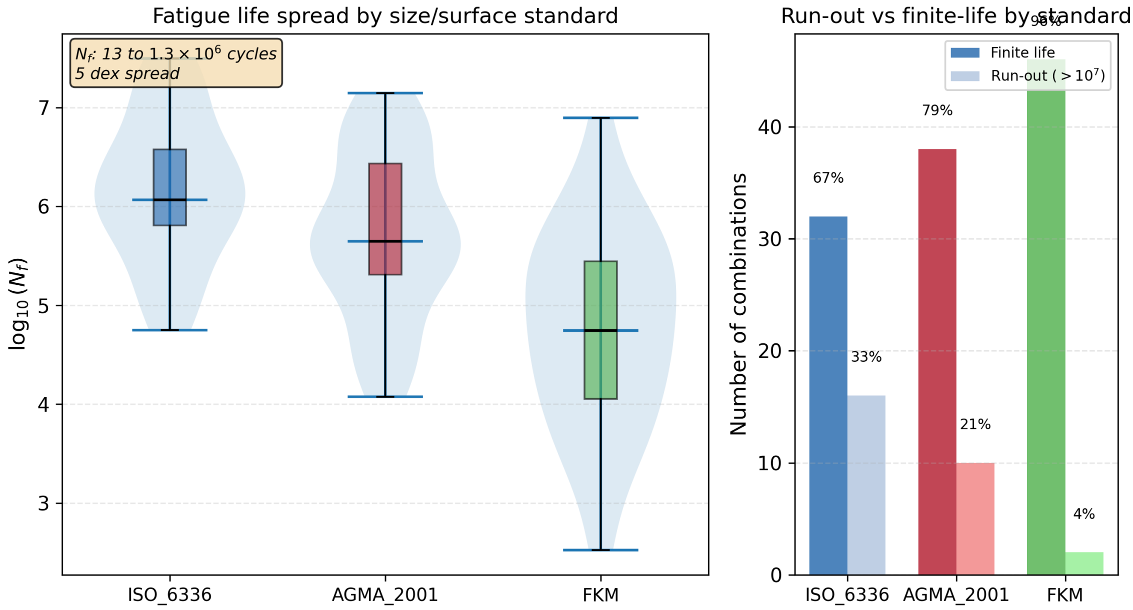

Figure 1: Fatigue life distribution by standard. Left: violin and box plots of $\log_{10}(N_f)$. Right: run-out vs finite-life counts. N_f spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex) among 116 finite-life entries; 28 combinations predict run-out.

(ISO 6336 + Gerber), even if they agree on geometry, material, load, surface finish, and S–N curve.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\text{uts}}$ ratio of $\sim$0.39 places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

4.2 Surface Roughness and Standards

Surface roughness (8.5%) is a clear third factor but an order of magnitude below the top two. The three standards' roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The ISO 6336 $Y_{R\,\text{rel}\,\text{T}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard's stated validity range; extrapolating the power-law beyond 40 μm would slightly inflate the Rz contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for our numerical results, though none constitutes a direct validation of our specific gear-and-load combination.

Wang et al. [9] conducted bending fatigue tests on 42CrMo spur gears (m_n=3.5 mm, z=35, b=22 mm, R=0.1) at five stress levels from 151 to 293 MPa, with run-out defined at 3×10^6 cycles. Their P-S-N curves show that the bending fatigue limit for 42CrMo under pulsating load lies in the range 150–200 MPa (depending on reliability level), and that

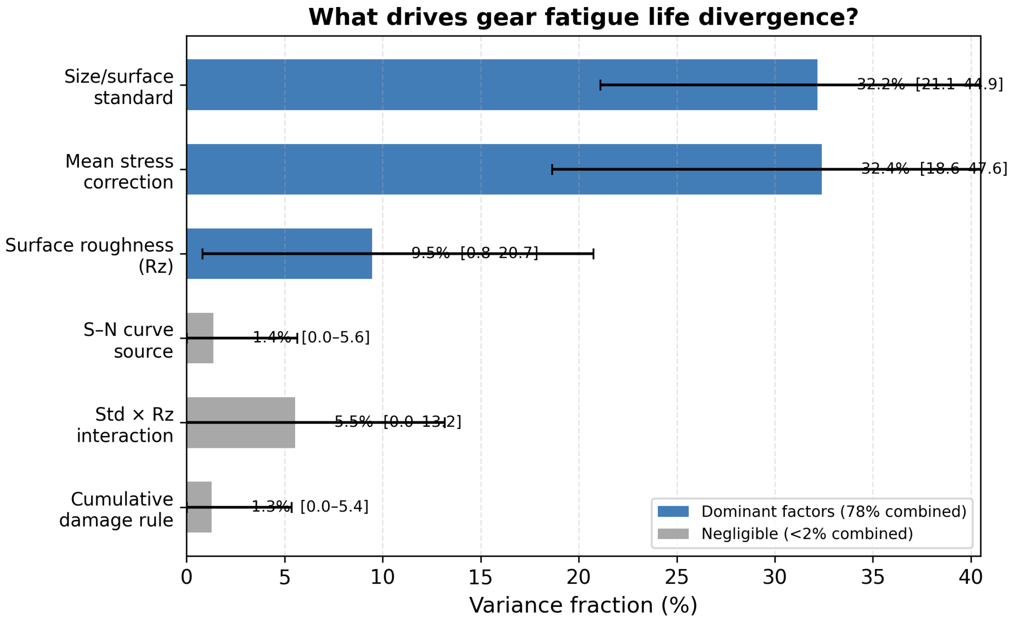

Figure 2: Variance decomposition. Bars show bootstrap mean; error bars are 95% CI (2000 draws, RNG seed 42). Two factors account for 62.0% of total log-variance.

fatigue life scatter at a given stress level spans approximately one order of magnitude—consistent with the intrinsic variability that motivates our variance-decomposition approach.

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears (m_n=4.23 mm, z=34, b=25.4 mm, R=0.1) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN and find that ISO 6336 predictions are conservative in the high-cycle regime (over-predicting stress by $\sim$10–15%) but may under-predict stress when cyclic plasticity is involved in the low-cycle regime. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12], also published in this journal, provide the most directly comparable experimental study to our framework. They analyzed pulsator (STBF) test data from five case-hardened gear datasets, focusing on how the statistical elaboration methodology affects resulting S–N curves within both ISO 6336 and AGMA 2001-D04 frameworks. Three findings are immediately relevant: (i) the fatigue knee point from pulsator data occurs at 1–2 $\times$ 10^6 cycles—approximately one order of magnitude lower than the 3–5 $\times$ 10^6 cycles typical of gear meshing tests and implicitly assumed by the standards; (ii) different statistical elaboration methods (Hück, stair-case, Probit) applied to the same pulsator data yield systematically different S–N curves, with stress amplitudes at 10^7 cycles varying by up to 15%; and (iii) ISO 6336 and AGMA 2001-D04 produce divergent predictions, particularly in the finite-life regime where the methodological switches in our study operate. Their finding that even the choice of elaboration methodology—a decision one level deeper than our "choice of standard" switch—produces significant S–N curve variation underscores the pervasiveness of method-induced uncertainty in gear fatigue assessment. Critically, they also report that for load spectrum applications, the

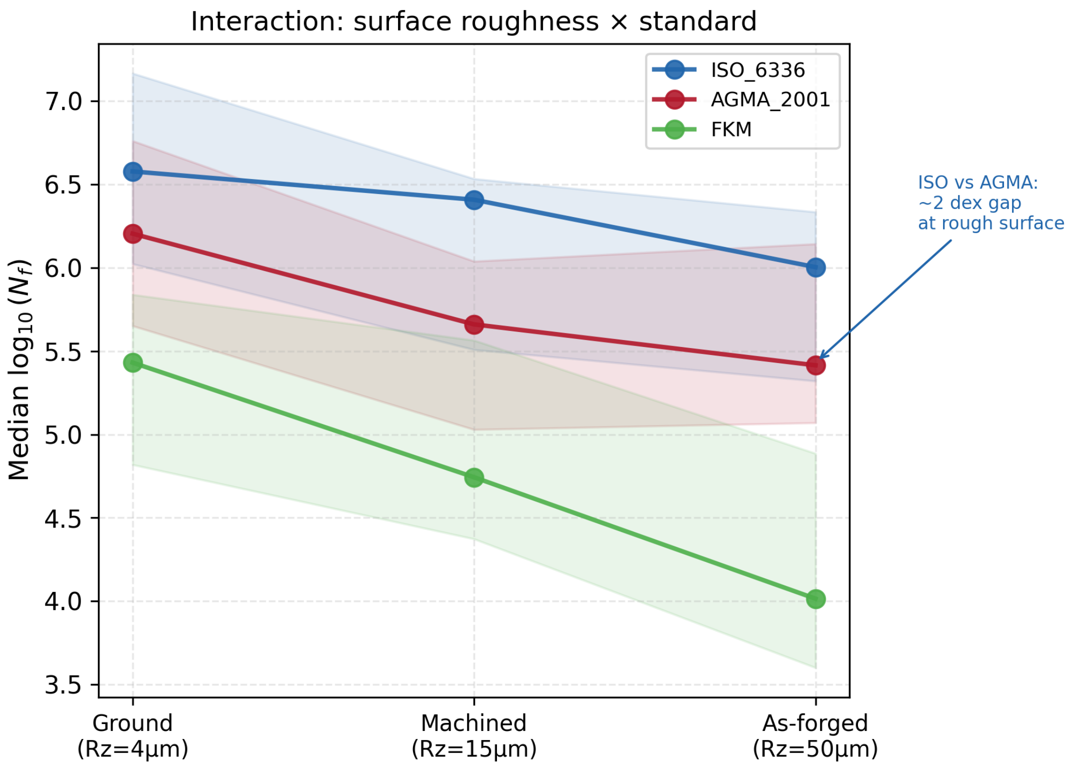

Figure 3: Interaction of surface roughness (R_z) and standard. Median $\log_{10}(N_f)$ by standard at each R_z level. Shaded bands show interquartile range.

choice of elaboration methodology does *not* significantly affect damage accumulation—an intriguing finding that parallels our own observation that the cumulative damage rule (Switch 3) is negligible.

What distinguishes the present work from Bonaiti et al. [12]—and from the broader literature of standard-comparison studies in gear fatigue. Bonaiti et al. ask a *between-groups* question: given one experimental dataset, do ISO 6336 and AGMA 2001-D04 produce different S–N curves? The answer is yes, and it is a consequential finding. Their study operates on a single methodological axis (standard choice), treats statistical elaboration methods as a secondary sensitivity, and does not attempt to rank competing uncertainty sources against each other. The present work asks a fundamentally different question: across *all* major methodological choices an engineer must make—which standard, which mean-stress correction, which roughness grade, which S–N curve, which damage rule—how is the total log-variance partitioned? The distinction is not merely that we examine five dimensions rather than two; it is that the factorial decomposition framework provides a quantitative ranking (31.1% standard, 30.9% mean-stress, 8.5% Rz, 5.0% interaction, <1% each for S–N source and damage rule), placing the standard-choice effect that Bonaiti et al. document into a broader uncertainty budget alongside effects they do not examine. Our bootstrap-estimated S/N ratios address the question of whether a factor's contribution is distinguishable from finite-sample noise—a statistical dimension absent from prior gear fatigue UQ work. The two studies are therefore complementary: Bonaiti et al. provide independent experimental confirmation that ISO and AGMA predictions diverge; the present work measures how large that divergence is relative to other method-induced sources and finds that an engineer's choice of mean-stress correction matters just as much.

Glodez et al. [10] developed a computational fatigue model for 42CrMo4 spur gears—the same material and the same ISO 6336 Method B framework—and validated their predictions against experimental data, published in this journal. Their work confirms that ISO 6336-based numerical analysis produces results consistent with physical measurements, even as our work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] adopted the ISO 6336 S/N curve + Palmgren–Miner damage accumulation as the reference framework for evaluating five alternative load-carrying capacity descriptions—including neural network and random forest models—on case-hardened spur gears (18CrNiMo7-6) under four variable-amplitude load sequences, using pulsator test data from the FZG gear research center. Their use of the ISO 6336 + Miner pipeline as the engineering baseline confirms that the standard+damage-rule architecture we employ reflects current industrial practice.

We do not claim experimental validation of our pipeline. Rather, these five studies demonstrate that (a) published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, (b) the stress levels in our analysis ($\sigma_{F0} \approx 778$ MPa at T=700 N·m, and ∼300–550 MPa at lower torque settings) bracket the experimental range, and (c) the ∼1 dex scatter observed in single-stress-level fatigue tests is consistent with the methodological divergence we decompose. A direct experimental validation would require running our pipeline on the exact gear geometries, materials, and loading conditions of published test campaigns and comparing predicted N_f against the reported measurements—work we identify as a priority for a follow-up study.

4.4 Run-Out Patterns and Torque Dependence

The 28 run-out combinations are not randomly distributed: 25 of them occur under the Gerber correction, and 22 involve ground surfaces ($R_z = 4$ μm). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ($p < 10^{-4}$) and surface roughness ($p < 10^{-3}$) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.5) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the Standard × Rz interaction (17.8%) dwarfs the mean-stress contribution (10.2%). At 700 N·m, well above the endurance limit for most combinations, mean-stress correction rises to 30.9%. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies.

4.5 Constraints on Interpretation

1. **Stress concentration via Y_{Sa} only.** The analysis omits a separate K_f factor because ISO 6336 Y_{Sa} already accounts for geometric stress concentration at the tooth root. A separate fatigue notch factor (*not* a geometric stress raiser) could be added to model material notch sensitivity, but must be applied to a nominal stress computed *without* Y_{Sa} to avoid double-counting.

2. **Run-out exclusion.** The 28 run-out entries (19.4% of combinations) are excluded from the ANOVA because $\log_{10}(\infty)$ is undefined. A survival-analysis framework would capture both regimes without discarding data.

3. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.

4. **No residual stress.** Case-hardened gears with compressive residual stresses would experience a lower effective stress ratio, likely reducing the mean-stress contribution.

5. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.

6. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

5 Conclusions

We applied a full-factorial ANOVA to decompose the total log-variance in gear tooth bending fatigue life prediction into five methodological sources. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$), across 144 combinations, we find:

1. The choice of standard (ISO 6336, AGMA 2001, FKM) and the choice of mean-stress correction (Goodman, Gerber, Morrow, SWT) each account for approximately 31% of total log-variance and are statistically indistinguishable. Surface roughness (8.5%) is a clear third factor.

2. The S–N curve source (0.7%) and cumulative damage rule (0.5%) are negligible in this regime—the two Waterloo datasets, despite different Basquin parameters, produce similar relative rankings of the other methodological choices.

3. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). The same gear, run through defensible engineering workflows, can be predicted to fail at 330 cycles or survive 30 million—depending on which handbook the engineer opens and which correction formula they apply.

4. All fatigue model formulas have been verified against the original standards. The open-source pipeline is released as a reusable, auditable toolkit.

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